

Date: 2026-06-30

Request ID: 100048706

Hotplate		
Analyte(s)	Ag	
SOP#	Determination of % Silver Composition	
Cocktail		
sample(s)	weight (g)	volume (mL)
200128410_1	0.5011	200
200128410_2	0.5051	200
200128411_1	0.5015	200
200128411_2	0.5015	200
200128412_1	0.5074	200
200128412_2	0.5057	200

Hotplate		
Analyte(s)	Pd	
SOP#	Determination of Palladium in Catalyst with or without Ag	
Cocktail		
sample(s)	weight (g)	volume (mL)
200128410_1	1.0033	200
200128410_2	1.0094	200
200128411_1	1.0026	200
200128411_2	1.0053	200
200128412_1	1.0023	200
200128412_2	1.0082	200

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LOI temp: 900 °C

sample	crucible (g)	crucible + sample (g)	after 900 °C	LOI (%)	correction
200128410	25.9901	28.0036	27.9845	0.95	0.9905

Ag ICP analysis

sample	Dilution	conc. [mg/L]	Ag_ppm [dried]	%Ag [dried]
200128410_1	10/9	0.834	373.40	0.04
200128410_2	10/9	0.856	380.21	0.04

ag result: 376.80 ppm

ag result: 0.04 %

ag lot: *Manufacturer: Assurance, 28-111AGY exp: 2027-02-28*

Pd ICP analysis

sample	Dilution	conc. [mg/L]	Pd_ppm [dried]	%Pd [dried]
200128410_1	10/9	1.333	298.08	0.03
200128410_2	10/9	1.209	268.71	0.03

pd result: 283.39 ppm

pd result: 0.03 %

pd lot: *Manufacturer: Ricca, 841005E exp: 2027-11-30*

Note(s):

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LOI temp: 900 °C

sample	crucible (g)	crucible + sample (g)	after 900 °C	LOI (%)	correction
200128411	27.1497	29.1583	29.1492	0.45	0.9955

Ag ICP analysis

sample	Dilution	conc. [mg/L]	Ag_ppm [dried]	%Ag [dried]
200128411_1	10/9	0.890	396.17	0.04
200128411_2	10/9	0.939	417.98	0.04

ag result: 407.07 ppm

ag result: 0.04 %

ag lot: *Manufacturer: Assurance, 28-111AGY exp: 2027-02-28*

Pd ICP analysis

sample	Dilution	conc. [mg/L]	Pd_ppm [dried]	%Pd [dried]
200128411_1	10/9	1.271	282.99	0.03
200128411_2	10/9	1.310	290.89	0.03

pd result: 286.94 ppm

pd result: 0.03 %

pd lot: *Manufacturer: Ricca, 841005E exp: 2027-11-30*

Note(s):

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LOI temp: 900 °C

sample	crucible (g)	crucible + sample (g)	after 900 °C	LOI (%)	correction
200128412	15.5573	17.5611	17.5474	0.68	0.9932

Ag ICP analysis

sample	Dilution	conc. [mg/L]	Ag_ppm [dried]	%Ag [dried]
200128412_1	10/9	0.826	364.25	0.04
200128412_2	10/9	0.847	374.76	0.04

ag result: 369.51 ppm

ag result: 0.04 %

ag lot: *Manufacturer: Assurance, 28-111AGY exp: 2027-02-28*

Pd ICP analysis

sample	Dilution	conc. [mg/L]	Pd_ppm [dried]	%Pd [dried]
200128412_1	10/9	1.260	281.28	0.03
200128412_2	10/9	1.324	293.84	0.03

pd result: 287.56 ppm

pd result: 0.03 %

pd lot: *Manufacturer: Ricca, 841005E exp: 2027-11-30*

Note(s):

$$\text{ppm } M+ = [\text{conc.}][\text{volume}][\text{dilution}]/[\text{weight}]$$

$$\%M+ = [\text{conc.}][\text{volume}][\text{dilution}]/([\text{weight}]*10,000)$$

$$\%MO = [\text{oxide factor}]*\%M+$$

A = crucible

B = crucible + sample

C = crucible + sample after drying

$$\%LOI = ([B-A]-[C-A])/(B-A)*100\%$$

$$\text{ppm } M+ \text{ [dried]} = \text{ppm } M+ / ([1-(\%LOI)/100])$$

$$\%M+ \text{ [dried]} = \%M+ / ([1-(\%LOI)/100])$$

$$\%MO \text{ [dried]} = \%MO / ([1-(\%LOI)/100])$$